Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/27/2023
PI Name:	Nusbaum, Derek M.D., Ph.D.		
Project Title:	Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Texas	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Principal Investigator (PI) is also affiliated with the University of Texas Medical Branch.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/22/2021	**End Date:**	01/14/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/14/2024 as Dr. Garbino took over the project in January 2024 (Ed., 4/18/24) NOTE: End date changed to 09/30/2024 per C. Ribeiro/JSC (Ed., 11/4/22)		
Key Personnel Changes/Previous PI:	NOTE: New PI (Alejandro Garbino, Ph.D.). for this investigation due to Dr. Nusbaum's departure from the laboratory conducting this research (Ed., 8/15/24). 2024 Report: Brett Siders was added to the study as a Co-I for his expertise in exercise physiology. Paul Smith was added to the study for his expertise in Virtual Reality programming/development.		
COI Name (Institution):	Abercromby, Andrew Ph.D. (NASA Johnson Space Center) Scully, Robert Ph.D. (KBR/NASA Johnson Space Center) Levine, Benjamin M.D. (University of Texas Southwestern Medical Center) Babb, Tony Ph.D. (University of Texas Southwestern Medical Center) Pawelczyk, James Ph.D. (Pennsylvania State University) Baughman, Alex M.S. (NASA Johnson Space Center) Bishop, Michael Ph.D. (KBR/NASA Johnson Space Center) Siders, Brett (KBR/NASA Johnson Space Center) Smith, Paul (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Carbon dioxide (CO2) may accumulate inside a space suit during extravehicular activities (EVA). The maximum inspired CO2 during a 1 hour contingency walkback scenario is unknown. The purpose of this study is to characterize the limit of acceptable performance (cognitive and physical) decrements and symptom severity for mission operations when subjected to elevated inspired CO2 levels (up to 30 mmHg). Specific Aim 1 – To quantify operational physical and cognitive performance across a range of inspired CO2 levels during simulated suited contingency EVA operations in a ground-based, shirtsleeve, laboratory study using standardized research techniques. Hypothesis 1.1. – Physical performance (ambulation speed and distance) of simulated EVA tasks will be reduced in a non-linear fashion with increasing partial pressures of inhaled carbon dioxide. Hypothesis 1.2. – Cognitive performance (visual search, spatial memory, paired associate learning) during simulated EVA tasks will be reduced in a non-linear fashion with increasing partial pressures of inhaled carbon dioxide. Specific Aim 2 – To qualify and quantify subjective metrics, including symptoms and self-assessment of performance experienced across a range of inspired CO2 levels during simulated suited contingency EVA operations in a ground-based, shirtsleeve, laboratory study using standardized research techniques. Hypothesis 2.1. – Subjects will experience carbon dioxide-associated symptoms when exposed to elevated partial pressures of inhaled carbon dioxide. Hypothesis 2.2. - Non-linear increases in symptom severity will be observed with increasing partial pressures of inhaled carbon dioxide. Hypothesis 2.3. – Self-assessment of performance will be impaired by increasing partial pressures of inhaled carbon dioxide, despite decrements in objective performance. Methods Study Design - This study will use a repeated measures ground-based study design. A hybrid reality EVA simulation environment (APACHE) will provide a physically and cognitively realistic simulation of a 1 hour contingency walkback, incorporating a passive treadmill and a fully immersive virtual reality simulation of lunar EVA. This study will be performed in a shirtsleeve environment, along with a simulated Hard Upper Torso, in a laboratory setting in Johnson Space Center's Building 21 with subjects breathing compressed air mixed with the prescribed fraction of CO2. Subjects will be asked to walk on a treadmill at a self-selected speed within a defined range, bounded by a minimum of 3 km/hr and a maximum speed associated with a maximum allowable metabolic intensity. Cognitive performance will be quantified using validated cognitive tests, as well as measures of functional task performance embedded within the APACHE environment. Physical performance will be measured by evaluating the distance translated by subjects during a one-hour simulated contingency walkback on a treadmill. The severity of specific, CO2 related symptoms (headache, dizziness, nausea, numbness, tingling, shortness of breath, air hunger, fatigue, euphoria) will be quantified via survey prior to the testing session, at 10 minute intervals during the testing session, and immediately following the testing session, as well as every 10 minutes for one hour post exposure to document recovery. This study will be conducted using a total of 15 different subjects. Each subject will be exposed to 7 different partial pressures of carbon dioxide (0, 5, 10, 15, 20, 25, 30 mmHg) for 1 hour at a time, each on a different day to provide sufficient time for washout of exposure. During the one hour of exposure, the subjects will breathe 21% oxygen compressed air (+CO2) via oronasal facemask. The order in which subjects are exposed to each of the different carbon dioxide partial pressures will utilize balanced randomization and the subjects will be blinded to the pressure of inspired carbon dioxide (PiCO2) levels. The range of CO2 levels to be used have been tolerated in previous studies.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research. The purpose of this study is to characterize the limit of acceptable performance (cognitive and physical) decrements and symptom severity for mission operations when subjected to elevated inspired CO2 levels. This study will determine the maximum inspired CO2 level that still enables crewmembers to complete a 1 hour contingency walkback within acceptable functional performance limits. Specifically, results will inform Exploration Extravehicular Mobility Unit (xEMU) suit design and operations and will be incorporated as a Contingency Inspired Carbon Dioxide Standard for Suited Operations in NASA-STD-3001. Furthermore, as this study will include the physiologic, symptomatic, and performance effects of CO2 at nominal partial pressures of PiCO2 (pressure of inspired carbon dioxide) in addition to contingency levels, this study will also provide data to inform and clarify overall risk of CO2 exposure in the flight environment, not just for suited, extravehicular activity (EVA) operations, which has been deemed an important human health and performance risk for future spaceflight missions. This is a rapid response study of operational relevance, informing future standards and requirements and thus driving future hardware design for future programs.
Research Impact/Earth Benefits:	The principal benefits of this study are expected to be in the area of operational space medicine, where these data will help inform future planetary EVA suit design and may drive EVA operational constraints. However, as this study deals with the cognitive and physical performance aspects of elevated CO2 exposure, it may have broader implications for clinical patients who are also exposed to elevated CO2, such as patients with chronic obstructive lung disease or patients with obesity hypoventilation syndrome.

Task Progress:	The fall of 2022 was spent evaluating how to incorporate a “suit simulator” into this study to ensure the physical and cognitive environment of the study mirrored the extravehicular activity (EVA) environment as much as possible. After a significant scientific review, it was determined that the highest yield for this study would be to incorporate a simulated Hard Upper Torso (HUT). Over the course of the fall, this HUT simulator was designed, fabricated, and tested, and procedures for its use were developed. In late fall 2022, we performed engineering runs to characterize the practice effect for the Digit Symbol Substitution Task (DSST), specifically within the treadmill and virtual reality (VR) environment. We also started test subject recruitment. We reached our 15+ subject recruitment target and started with peak cycle testing for baseline data collection. We also started with HUT sizing and treadmill and DSST familiarization testing for the subjects at the same time. Winter 2022/2023 was spent working on a redesign of the breathing system to improve its utility and functionality. This included the incorporation of an air compressor cart as the source of compressed air for breathing which went through its own refurbishment, testing. and Test Readiness Review (TRR) in April. This study has now been through a dTRR at the beginning of April to certify the new breathing system and the HUT simulator. We are now in a posture to proceed with data collection imminently. Data collection will begin when able, so as to ensure we are deconflicted from other studies and to ensure sufficient personnel availability. NOTE: Continued by "Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario" (PI: Garbino). See project with same title under Dr. Garbino for later reporting.
Bibliography Type:	Description: (Last Updated:)